HotSense 2026: The 2nd Workshop on Thermal Sensing and Computing

Workshop Proposal for ACM UbiComp / ISWC 2026

Organizers:
Chenshu Wu (Assistant Professor, The University of Hong Kong);
Fanglin Bao (Assistant Professor, Westlake University);
Mayank Goel (Associate Professor, Carnegie Mellon University);
Petteri Nurmi (Professor, University of Helsinki)
,

1 TOPIC, RATIONALE, AND OBJECTIVES

Thermal sensing is emerging as a practical, privacy-aware sensing modality for ubiquitous and wearable computing. HotSense 2026 aims to gather the growing community around thermal sensing and computing to share advances, identify open challenges, and accelerate real-world impact.

Topic: HotSense 2026 is the 2nd Workshop on Thermal Sensing and Computing, focusing on the full stack of thermal-centric ubiquitous and wearable systems—from sensing hardware and calibration to algorithms, deployment, and human-facing applications.

Motivation: Thermal sensing is emerging as a powerful and timely modality for ubiquitous and wearable computing, especially with the advent of low-cost, low-resolution thermal infrared arrays that can capture heat signatures of people and objects without recording identifiable visual details. In many real-world deployments, visible-spectrum cameras provide rich sensing fidelity but introduce significant privacy concerns, while RF sensing (*e.g.*, Wi-Fi or radar) can better preserve privacy but often provides weaker spatial structure for fine-grained perception. Thermal sensing offers a compelling middle ground: it can deliver spatially meaningful signals for activity and environment understanding while inherently limiting sensitive visual content. Recent systems have demonstrated that even a single commodity thermal array can support advanced tasks such as multi-person detection and ranging by lifting 2D heat measurements into 3D structure.

Why now?: Thermal hardware is rapidly becoming more accessible and integrable (*e.g.*, compact LWIR modules, Melexis microbolometer arrays, and industry-grade thermal sensors), and the research momentum is growing across mobile systems, HCI, computer vision, robotics, and healthcare. This creates a timely opportunity for HotSense 2026 to consolidate a still-fragmented community, share best practices and open challenges, and shape a shared research agenda for thermal sensing and computing.

Goals: The primary goal of HotSense 2026 is to build and nurture a community around thermal sensing and computing within ubiquitous and wearable systems research. We aim to achieve this goal through the following outcomes-oriented objectives:

- *Build an interdisciplinary forum:* bring together researchers across ubiquitous computing, mobile systems, HCI, computer vision, robotics, sensing hardware/physics, and healthcare to share results and brainstorm new ideas leveraging thermal sensors.
- *Accelerate innovation and deployment:* surface end-to-end lessons from real systems (from calibration and sensing pipelines to on-device inference and evaluation) that reduce barriers to practical deployment.
- *Identify shared challenges and best practices:* discuss recurring "hard problems" (noise, emissivity variation, occlusion, domain shift, evaluation) and compare methodological choices across sub-communities.
- *Bridge data-driven and physics-informed approaches:* encourage models that integrate learning with heat-transfer and optics constraints for robustness and interpretability.
- *Seed community artifacts:* catalyze datasets, benchmarks, and evaluation protocols, and produce a concise public workshop summary capturing key takeaways and open problems.

Topics of Interest (including but not limited to):

- Human sensing (e.g., gesture, activity, and pose recognition) using thermal/infrared data
- Thermal localization and tracking
- Thermal depth estimation and ranging
- Thermal imaging and space reconstruction
- Robotic perception and navigation augmented by thermal vision
- Material sensing and object classification using thermal signatures
- Privacy-preserving sensing for homes, healthcare, and interactions

- Thermal sensing for building occupancy and energy efficiency
- Thermal generative models and synthesis
- Multimodal learning and foundation models for thermal perception
- Lower-power and embedded systems for thermal sensing
- Large-scale and real-world thermal sensing systems
- Novel sensors, optics, and metasurface designs
- Novel applications in health, AR/VR, cities, emergency response, low-altitude, etc

2 ORGANIZERS

Chenshu Wu (https://www.cswu.me/), The University of Hong Kong, HK SAR, China:
Chenshu Wu is an Assistant Professor in the Department of Computer Science at the University of Hong Kong, where he leads the HKU AIoT Lab. He received his B.E. and Ph.D. degrees both from Tsinghua University. He served as the Chief Scientist of Origin Wireless. His research focuses on wireless and mobile AIoT. He has published 3 books and holds 90+ filed/granted US and international patents. His research on WiFi sensing has been deployed as award-winning products worldwide. He is the recipient of IEEE Communications Society Fred W. Ellersick Prize, ICEdge AIoT Rising Star Award, NSFC Excellent Young Scientists Fund (HK & Macau), NAM Healthy Longevity Catalyst Award, and CCF Outstanding Doctoral Dissertation Award. He is currently an associate editor of ACM IMWUT and ACM TIoT. He was Journal Track Co-Chair of EWSN 2025, and TPC Co-Chair of IEEE ICPADS 2024, and served on the TPC of MobiSys, SenSys, MobiHoc, ICDCS, INFOCOM, etc. He also served on the organizing committee of CoNEXT 2025, RFCom 2024, CPS-IoT Week 2024, SenSys 2023, APSys 2021, and he co-organized the Wireless AI Perception workshop at CVPR'22 and CPD workshop at UbiComp'21. He founded and served as one of the chairs of the inaugural HotSense workshop (co-located with ACM MobiCom 2025). Dr. Wu co-authored the recent TADAR and TAPOR systems, demonstrating the potential of low-resolution thermal arrays for privacy-preserving human sensing.

Fanglin Bao (https://en.westlake.edu.cn/faculty/fanglin-bao.html), Westlake University, China:
Dr. Fanglin Bao is an assistant professor in the School of Science, Westlake University, focusing on information physics and optics. Dr. Bao received his B.S. in Physics (2011) and Ph.D. in Optics (2016) from Zhejiang University. He studied quantum vacuum fluctuations, worked out the renormalization of the Casimir force in inhomogeneous systems, and discovered the inhomogeneity-induced Casimir transport phenomenon — a nanoscale 'maglev train' based on quantum levitation. Before joining Westlake University in 2024, he was a postdoctoral scholar and later a research scientist at Purdue University, where he worked on quantum optical sensing. He developed heat-assisted detection and ranging (HADAR), quantum-accelerated imaging, and adaptive photon-thresholding detection. For the series of work on HADAR, Dr. Bao was recognized with awards, including the 2023 Intelligent Computing Innovator Award from MIT Technology Review.

Mayank Goel (http://www.mayankgoel.com/), Carnegie Mellon University, USA:
Mayank Goel is an Associate Professor in the Software & Societal Systems Department (S3D) and the Human-Computer Interaction Institute (HCII) in the School of Computer Science at CMU, where he leads the Smash Lab. His research interests span Human-Computer Interaction, ubiquitous computing, sensing, signal processing, and applications of machine learning, with an emphasis on building practical and deployable sensing systems for health, global development, and novel interaction. He regularly collaborates with mechanical and biomedical engineers as well as doctors, nurses, and community health workers; some of his inventions have been deployed in clinics around the world and licensed to multiple companies. He received his Ph.D. in Computer Science and Engineering from the University of Washington (2016), M.S. in Computer Science from Georgia Institute of Technology (2009), and B.Tech. in CSE from GGS Indraprastha University, India (2007). He is a recipient of a Google Faculty Research Award and a Microsoft Research Ph.D. Fellowship, and has received multiple best paper nominations and honorable mention awards at CHI, UbiComp, UIST, and PerCom. He has served in major conference leadership roles including General Conference Chair of UIST 2024, Gadgets Chair of UbiComp 2017, and Workshops and Tutorials Chair of UbiComp 2019, and served on the IMWUT editorial board (2017–2023).

Petteri Nurmi (https://researchportal.helsinki.fi/en/persons/petteri-nurmi/), University of Helsinki, Finland:
Petteri Nurmi is an internationally well-recognized researcher in the fields of ubiquitous computing and sensing. He is a Professor at the University of Helsinki. Prof. Nurmi has a comprehensive publication record, with over 180 published articles, and an extensive citation record (5900+ citations, h-index of 41, i10-index of 114). He consistently publishes at top-ranked conferences, such as UbiComp, MobiCom, SenSys, MobiSys, IUI, WWW, and PerCom, and journals such as IEEE Pervasive, IEEE Transactions on Mobile Computing, Pervasive and Mobile Computing, and ACM Transactions on Intelligent Information Systems. He has served as an external evaluator for ERC starting grant proposals; he was the General Co-Chair of ACM MobiSys 2023, Program Chair for MobiCASE 2015; and he serves as a program committee member and reviewer for several top conferences in the field.

He has also actively participated in conference organization activities (workshop chair for PerCom 2018 and Pervasive 2009, as Demonstration Chair for PerCom 2017, and as poster chair for MobileHCI 2013). He also formerly worked as Science Advisor for Moprim, a company focusing on analysis of mobility information.

3 VENUE FIT AND IMPACT

Building on the success of the inaugural HotSense workshop co-located with ACM MobiCom 2025, HotSense 2026 is timely for UbiComp/ISWC because thermal sensing directly supports core conference priorities: privacy-aware sensing in homes and healthcare; always-on embedded perception; wearable and on-body thermal modalities; and multi-modal fusion for robust context inference in the wild.

Why UbiComp/ISWC (vs. MobiCom) this year: While MobiCom is an excellent venue for mobile networking and systems, UbiComp/ISWC offers a uniquely cross-cutting community spanning in-situ sensing, human-centered and wearable computing, privacy/ethics, and real-world deployments. Moving HotSense to UbiComp/ISWC 2026 broadens participation beyond the networking community and better aligns with the workshop's goal of connecting thermal sensing research across systems, HCI, vision, and health.

Impact to the community: HotSense 2026 will make impact by (i) consolidating a sustained community around thermal sensing and computing, (ii) enabling high-signal exchange between academia and industry (where low-cost thermal hardware constraints are most visible), and (iii) producing a short public workshop summary capturing key takeaways, open problems, and candidate benchmark directions to guide subsequent research.

4 ATTENDANCE AND PARTICIPATION

Expected attendance: Based on the momentum of the inaugural HotSense workshop and growing interest in privacy-preserving and in-situ sensing at UbiComp/ISWC, we expect to receive on the order of 15–30 paper submissions in response to our call for papers and approximately *30–50 participants*. Participants will likely include researchers and practitioners from academia and industry. We also foresee interest from UbiComp/ISWC main conference attendees who are curious about exploring thermal sensing as a new research direction.

Open participation: The workshop will be *open* to all UbiComp/ISWC attendees. Participants without accepted submissions can still join keynotes, discussions, and interactive sessions; accepted authors will receive priority for talk/poster slots.

5 SUBMISSIONS

Lightweight submissions: To keep the workshop lightweight and discussion-oriented (rather than a mini-conference), we solicit a single paper type with simple requirements. (i) Submissions must be original, unpublished work, and not under review elsewhere. (ii) Papers should be submitted in PDF format, be no longer than *5 pages excluding references*, and follow the UbiComp/ISWC (IMWUT) templates and formatting instructions.

Review and selection process: Submissions will undergo a *single-round, lightweight review* focusing on relevance to the workshop, novelty/insight, discussion value, and clarity. Each submission will receive at least *two reviews*. We will keep the program selective enough to ensure high-quality discussion, while avoiding a heavy multi-round process. We anticipate a *small, focused program committee* (PC) spanning systems, HCI, vision, and thermal sensing; the PC will be announced on the workshop website.

Presentation formats: To prioritize interaction, we will *limit the number of long talks* and use a mix of:

- *Lightning talks* for accepted papers (5 min talk + 2 min Q&A each), and
- *Posters* for all accepted papers to enable deeper discussion and hands-on interaction.

We will reserve a small number of slots for *invited* talks that broaden the perspective beyond accepted submissions.

Publication: We plan to publish accepted papers in the *UbiComp/ISWC Adjunct Proceedings and ACM Digital Library*, following the UbiComp/ISWC 2026 templates and formatting instructions.

6 PROGRAM COMMITTEE

Following UbiComp/ISWC guidance that workshops should remain lightweight (and not become mini-conferences), HotSense 2026 will use a *small, focused* program committee rather than a large PC. The PC will cover complementary expertise across ubiquitous computing, mobile/wearable sensing, HCI, computer vision/robotics, and thermal sensing hardware/physics, and will support a single-round, discussion-oriented review process (at least two reviews per submission).

Tentative members: A tentative list of PC members will be finalized and announced on the workshop website after the proposal decision.

7 WORKSHOP PLAN

The HotSense 2026 workshop is designed to maximize interaction, collaboration, and actionable outcomes for the thermal sensing community. Below, we outline the planned structure, logistics, and unique features that will shape the workshop experience.

Pre-workshop Preparation: We will maintain a dedicated website (continuing the `hotsense.github.io` domain) to host the CFP, committee, submission instructions, and the final program. We will promote the workshop via community channels (UbiComp/ISWC and SIGMOBILE mailing lists, relevant research group networks), social media outreach on LinkedIn, and direct outreach to authors, speakers, and attendees from the previous HotSense workshop, alongside targeted outreach to researchers and industry working on thermal imaging hardware and low-cost thermal arrays.

Workshop Activities (Half-Day Schedule): We propose a *half-day* (approximately 4 hours) program, designed to balance invited content, paper presentations, and structured discussions. The exact start time will follow the UbiComp/ISWC workshop schedule; below is a representative half-day agenda:

- **08:30 – 08:40:** Opening Remarks, Goals, and Format
- **08:40 – 09:30:** Keynote I
- **09:30 – 10:30:** Lightning Talks I (accepted papers; 5 min talk + 2 min Q&A each)
- **10:30 – 10:45:** Coffee Break + Poster/Demo Setup
- **10:45 – 11:30:** Keynote II or Industry Talk
- **11:30 – 12:00:** Interactive Poster/Demo Session
- **12:00 – 12:30:** Closing Remarks and Community Building

We will prioritize substantial discussion time by limiting long single-paper talks and using lightning talks plus posters to maximize interaction.